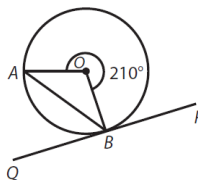


# Year 9

|    |                                                                                                                                                                                                                                                                                                                                                      |                                                                                      |  |  |  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--|--|--|
| 1  | Find the last digit of $549^{123} \times 549^{123}$                                                                                                                                                                                                                                                                                                  |                                                                                      |  |  |  |
| 2  | Find the number of zeros at the end of $17!$                                                                                                                                                                                                                                                                                                         |                                                                                      |  |  |  |
| 3  | How many positive integers $n$ satisfy the following condition?<br>$(130n)^{50} > n^{100} > 2^{200}$                                                                                                                                                                                                                                                 |                                                                                      |  |  |  |
| 4  | If $f(x) = ax + b$ & $f(f(f(x))) = 64x + 63$ , what is the sum of a & b?                                                                                                                                                                                                                                                                             |                                                                                      |  |  |  |
| 5  | In a certain year the price of gasoline rose by 20% during January, fell by 20% during February, rose by 25% during March, and fell by $x\%$ during April. The price of gasoline at the end of April was the same as it had been at the beginning of January. To the nearest integer, what is $x$ ?                                                  |                                                                                      |  |  |  |
| 6  | In a specific week this year, the probability of raining on any given day is 50%. If the probability of raining on exactly 2 days in that week is given in the form $\frac{a}{2^7}$ , what is the value of a?                                                                                                                                        |                                                                                      |  |  |  |
| 7  | Jacob wanted to get rid of most of his toy cars. He gave one-sixth of his collection, plus 8 cars, to Xu, and one-fourth of what was left, plus 4 cars, to Olivia. Then he gave half of what was left to Ahmed. Then he gave 40% of what was left, plus 5 cars, to Jim. He kept the last 10 cars. How many toy cars did Jacob have at the beginning? |                                                                                      |  |  |  |
| 8  | Let $n$ be the largest integer that is the product of exactly 3 distinct prime numbers $d$ , $e$ , and $10d + e$ , where $d$ and $e$ are single digits. What is the sum of the digits of $n$ ?                                                                                                                                                       |                                                                                      |  |  |  |
| 9  | The average of the number list $\{1,2,3, \dots, 98,99, x\}$ is $100x$ . $x$ can be expressed in the form of $\frac{m}{n}$ , where m and n are coprime integers, find $m + n$ .                                                                                                                                                                       |                                                                                      |  |  |  |
| 10 | How many right triangles have integer leg lengths $a$ and $b$ and a hypotenuse of length $b + 1$ , where $b < 100$ ?                                                                                                                                                                                                                                 |                                                                                      |  |  |  |
| 11 | If $AB$ is a chord of the circle in the diagram, & $PQ$ is a tangent to the circle at point B, find the value of $\angle ABQ$ (in degrees)                                                                                                                                                                                                           |  |  |  |  |
| 12 | If the mean deviation of the numbers $1, 1 + d, 1 + 2d, \dots, 1 + 100d$ from their mean is 255, then $100d$ is equal to?                                                                                                                                                                                                                            |                                                                                      |  |  |  |
| 13 | Keith had a birthday on Tuesday, May 27 in the leap year 2008. In what year will his birthday next fall on a Saturday? Write the last 2 digits of the year.                                                                                                                                                                                          |                                                                                      |  |  |  |
| 14 | Let $S_n$ denotes the sum of first n terms of an arithmetic series. If $S_{2n} = 3S_n$ , then the value of $\frac{S_{3n}}{S_n}$ is equal to?                                                                                                                                                                                                         |                                                                                      |  |  |  |

15 Numbers  $x$  &  $y$  are positive integers that fulfil the equation

$$\frac{6}{x} + \frac{7}{y} = 1$$

Find the largest value of  $x + y$

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16 On a 50-question multiple choice math contest, students receive 4 points for a correct answer, 0 points for an answer left blank, and  $-1$  point for an incorrect answer. Larisse's total score on the contest was 99. What is the maximum number of questions that Larisse could have answered correctly?

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17 Quadrilateral  $ABCD$  has  $AB = BC = CD$ ,  $\angle ABC = 70^\circ$  and  $\angle BCD = 170^\circ$ . What is  $\angle BAD$  in degrees?

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18 Thirty identical marbles are to be distributed to 6 boys. Each boy gets at least 3 marbles. How many ways are there to distribute the marbles to the boys?

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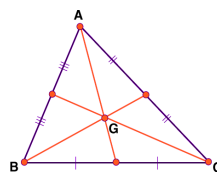
19 Three red beads, two white beads, and one blue bead are placed in line in random order. The probability that no two neighbouring beads are the same color can be written in the form of  $\frac{100}{a}$ , find  $a$ .

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20 Given a triangle  $ABC$ , let  $G$  be the centroid\* of the triangle and  $l$  be a line not intersecting any of the sides of the triangle. If the distances of the vertices  $A$ ,  $B$  and  $C$  to the line  $l$  are 5, 10 and 12 respectively, what is the distance from  $G$  to  $l$ ?

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*\*The centroid of a triangle is the intersection point of its three medians, which are lines connecting each vertex to the midpoint of the opposite side. Refer to the diagram as an example, but keep in mind that the diagram is NOT related to the question.*



## Year 9

| Question | Correct Answer |
|----------|----------------|
| 1        | <i>1</i>       |
| 2        | <i>3</i>       |
| 3        | <i>125</i>     |
| 4        | <i>7</i>       |
| 5        | <i>17</i>      |
| 6        | <i>21</i>      |
| 7        | <i>96</i>      |
| 8        | <i>12</i>      |
| 9        | <i>151</i>     |
| 10       | <i>6</i>       |
| 11       | <i>75</i>      |
| 12       | <i>1010</i>    |
| 13       | <i>17</i>      |
| 14       | <i>6</i>       |
| 15       | <i>56</i>      |
| 16       | <i>29</i>      |
| 17       | <i>85</i>      |
| 18       | <i>6188</i>    |
| 19       | <i>600</i>     |
| 20       | <i>9</i>       |

# Year 10

|    |                                                                                                                                                                                                                                                                                                            |  |  |  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
| 1  | Find the number of zeros at the end of $17!$                                                                                                                                                                                                                                                               |  |  |  |
| 2  | How many positive integers $n$ satisfy the following condition?<br>$(130n)^{50} > n^{100} > 2^{200}$                                                                                                                                                                                                       |  |  |  |
| 3  | The base 6 expansion of one number is 550. The base 5 expansion of another number is 3440. What is their greatest common divisor expanded in base 4?                                                                                                                                                       |  |  |  |
| 4  | Two trains leave a railway station at the same time. The first train travels due West and the second train due North. The first train travels 5 km/hr faster than the second train. If after two hours, they are 50 km apart, then find the sum of the average speeds of the 2 trains.                     |  |  |  |
| 5  | What is the remainder when $3^0 + 3^1 + 3^2 + \dots + 3^{2026}$ is divided by 8?                                                                                                                                                                                                                           |  |  |  |
| 6  | What is the sum of the last three digits of $2011^5$ ?                                                                                                                                                                                                                                                     |  |  |  |
| 7  | On a 50-question multiple choice math contest, students receive 4 points for a correct answer, 0 points for an answer left blank, and $-1$ point for an incorrect answer. Larisse's total score on the contest was 99. What is the maximum number of questions that Larisse could have answered correctly? |  |  |  |
| 8  | Quadrilateral $ABCD$ has $AB = BC = CD$ , $\angle ABC = 70^\circ$ and $\angle BCD = 170^\circ$ . What is $\angle BAD$ in degrees?                                                                                                                                                                          |  |  |  |
| 9  | The expression $(x^2 + 1)(y^2 + 1) = 6(x + y) - 9$ . Find the value of $x^2 + y^2$ .                                                                                                                                                                                                                       |  |  |  |
| 10 | The length of the interval of solutions of the inequality $a \leq 2x + 3 \leq b$ is 10. What is $b - a$ ?                                                                                                                                                                                                  |  |  |  |
| 11 | The value of $\sqrt{18} + \sqrt{27}$ is between 2 consecutive natural numbers. Find the sum of the pair of numbers.                                                                                                                                                                                        |  |  |  |
| 12 | Vikas can use <b>all</b> the money he has to buy either one less or one more of a certain number of jeans. If that certain number of jeans is $x$ , and Vikas has $\$(x^3 + 2ax + b)$ , how much money does Vikas have left after buying 4 jeans?                                                          |  |  |  |
| 13 | Yesterday Hendrik drove 1 hour longer than Ian at an average speed 5 miles per hour faster than Ian. Jonathan drove 2 hours longer than Ian at an average speed 10 miles per hour faster than Ian. Hendrik drove 70 miles more than Ian. How many more miles did Jonathan drive than Ian?                  |  |  |  |

- 14 Let  $f(x)$  be a quadratic polynomial with real coefficients satisfying the inequality:

$$x^2 - 2x + 2 \leq f(x) \leq 2x^2 - 4x + 3$$

for all real numbers  $x$ . If  $f(11) = 181$ , find  $f(16)$ .

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- 15 Let  $k = 2026^2 + 2^{2026}$ . What is the units digit of  $k^2 + 2^k$ ?

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- 16 Point B is on AC with  $AB = 9$  and  $BC = 21$ . Point D is not on AC so that  $AD = DC$ , and AD and BD are integers. Find the sum of all possible perimeters of  $\triangle ACD$ .

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- 17 Positive integers  $a, b, c$ , &  $d$  satisfy the relation  $a > b > c > d$ , and also the equations  $a + b + c + d = 2010$  &  $a^2 - b^2 + c^2 - d^2 = 2010$ . Find the number of possible values of  $a$ .

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- 18 The number 4 can be expressed as the sum of the digits 1 or 2 in five different ways as shown below.

- $4 = 1 + 1 + 1 + 1$
- $4 = 1 + 1 + 2$
- $4 = 1 + 2 + 1$
- $4 = 2 + 1 + 1$
- $4 = 2 + 2$

How many ways can the number 9 be expressed as the sum of the digits 1 or 2?

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- 19 Suppose that  $x, y, z$  are complex numbers such that  $xy = -80 - 320i$ ,  $yz = 60$ , and  $xz = -96 + 24i$ , where  $i = \sqrt{-1}$ . Then there are real numbers  $a$  and  $b$  such that  $x + y + z = a + bi$ . Find  $a^2 + b^2$ .

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- 20 The entries in a  $3 \times 3$  array include all the digits from 1 through 9, arranged so that the entries in every row and column are in increasing order. How many such arrays are there?

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## Year 10

| Question | Correct Answer |
|----------|----------------|
| 1        | <i>3</i>       |
| 2        | <i>125</i>     |
| 3        | <i>33</i>      |
| 4        | <i>35</i>      |
| 5        | <i>5</i>       |
| 6        | <i>6</i>       |
| 7        | <i>29</i>      |
| 8        | <i>85</i>      |
| 9        | <i>7</i>       |
| 10       | <i>20</i>      |
| 11       | <i>19</i>      |
| 12       | <i>60</i>      |
| 13       | <i>150</i>     |
| 14       | <i>406</i>     |
| 15       | <i>6</i>       |
| 16       | <i>380</i>     |
| 17       | <i>501</i>     |
| 18       | <i>55</i>      |
| 19       | <i>74</i>      |
| 20       | <i>42</i>      |

# VCE

1 Jacob wanted to get rid of most of his toy cars. He gave one-sixth of his collection, plus 8 cars, to Xu, and one-fourth of what was left, plus 4 cars, to Olivia. Then he gave half of what was left to Ahmed. Then he gave 40% of what was left, plus 5 cars, to Jim. He kept the last 10 cars. How many toy cars did Jacob have at the beginning?

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4 Let  $S_n$  denotes the sum of first  $n$  terms of an arithmetic series. If  $S_{2n} = 3S_n$ , then the value of  $\frac{S_{3n}}{S_n}$  is equal to?

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Find the largest value of  $x + y$

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8 The expression  $(x^2 + 1)(y^2 + 1) = 6(x + y) - 9$ . Find the value of  $x^2 + y^2$

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9 The length of the interval of solutions of the inequality  $a \leq 2x + 3 \leq b$  is 10. What is  $b - a$ ?

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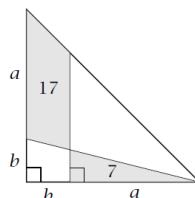
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|  |  |  |
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- 13 Three red beads, two white beads, and one blue bead are placed in line in random order. The probability that no two neighbouring beads are the same color can be written in the form of  $\frac{100}{a}$ , find  $a$ .

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- 14 Two shaded areas in the right triangle are 17 and 7, as marked. Find the value of  $ab$ .



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- 15 What is the least number of weights required to weigh any object of an integer number of grams from 1 to 35 grams? The weights themselves must also weigh an integer number of grams.

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- 16 Let  $S$  be a subset of  $\{1, 2, 3, \dots, 30\}$ . The subset  $S$  has the property that no pairs of distinct elements that has a sum divisible by 5. What is the largest size of  $S$ ?

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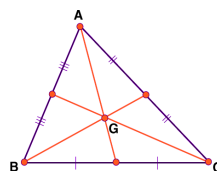
- 17 For how many ordered triples  $(x, y, z)$  of nonnegative integers less than 20 are there exactly two distinct elements in the set  $\{i^x, (1 + i)^y, z\}$ , where  $i = \sqrt{-1}$ ?

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- 18 Given a triangle  $ABC$ , let  $G$  be the centroid\* of the triangle and  $l$  be a line not intersecting any of the sides of the triangle. If the distances of the vertices  $A$ ,  $B$  and  $C$  to the line  $l$  are 5, 10 and 12 respectively, what is the distance from  $G$  to  $l$ ?

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\*The centroid of a triangle is the intersection point of its three medians, which are lines connecting each vertex to the midpoint of the opposite side. Refer to the diagram as an example, but keep in mind that the diagram is NOT related to the question.



- 19 In a  $6 \times 5$  grid, how many ways are there to form a set of three distinct items such that no two selected items are in the same row or column?

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- 20 Suppose that  $x, y, z$  are complex numbers such that  $xy = -80 - 320i$ ,  $yz = 60$ , and  $xz = -96 + 24i$ , where  $i = \sqrt{-1}$ . Then there are real numbers  $a$  and  $b$  such that  $x + y + z = a + bi$ . Find  $a^2 + b^2$ .

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## VCE

| Question | Correct Answer |
|----------|----------------|
| 1        | <i>96</i>      |
| 2        | <i>17</i>      |
| 3        | <i>17</i>      |
| 4        | <i>6</i>       |
| 5        | <i>56</i>      |
| 6        | <i>29</i>      |
| 7        | <i>85</i>      |
| 8        | <i>7</i>       |
| 9        | <i>20</i>      |
| 10       | <i>501</i>     |
| 11       | <i>55</i>      |
| 12       | <i>6188</i>    |
| 13       | <i>600</i>     |
| 14       | <i>20</i>      |
| 15       | <i>4</i>       |
| 16       | <i>13</i>      |
| 17       | <i>225</i>     |
| 18       | <i>9</i>       |
| 19       | <i>1200</i>    |
| 20       | <i>74</i>      |